

Software Requirements Specification (SRS) Template

Project: Railway Ticket Reservation System

Version: 1.0

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Date: 06-09-2026

Status: Submitted

Revision history

Version	Date	Author	Change summary	Approval
1.0	06-09-2026	Instructor	SRS with diagrams embedded	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for a Railway Ticket Reservation System. It defines functional and non-functional requirements, including train searching, ticket booking, ticket cancellation, passenger management, seat allocation, and booking record management.

1.2 Scope

The system allows users to search for trains, view train details and seat availability, book railway tickets, view existing bookings, and cancel tickets using a unique PNR number. The system maintains train, passenger, seat, and booking information using file-based storage. The system is intended as a console-based C++ application.

1.3 Audience

This SRS is intended for the project developers, testers, instructors, and evaluators of the Railway Ticket Reservation System.

1.4 Definitions

TERM	MEANING
PNR	Passenger Name Record/ unique booking identifier
Train number	Unique number assigned to each train
Seat number	Unique number assigned to each passenger after booking
Booking	Process/ status of reserving a seat
Cancellation	Process of cancelling an existing ticket
Fare	Price of ticket

2. Overall description

2.1 Product perspective

The Railway Ticket Reservation System is a standalone console-based application which will be developed in C++. It provides a simple interface for searching trains, checking seats availability, booking tickets, viewing booking details, and cancelling tickets. The system stores train and reservation information using local files and updates seat availability accordingly whenever a booking or cancellation is performed.

2.2 Major product functions (detailed)

- User menu and navigation
- Display available trains
- Search trains by source and destination
- Search trains by train number
- View train details
- Check seat availability
- Book a ticket
- Enter and validate passenger information
- Generate unique PNR
- Assign available seat
- Calculate ticket fare
- Display booking confirmation
- Search booking using PNR
- View booking details
- Cancel ticket
- Update seat availability after cancellation
- Store booking information in files
- Load booking information when the program starts

2.3 User roles and characteristics (expanded)

- **Passenger/User:** Searches trains, checks availability, books tickets, views bookings, and cancels tickets.
- **Administrator:** Logs in, manages train information, and can view booking records.

2.4 Operating environment

The system will run as a C++ console application on a Windows/Linux computer using a standard C++ compiler such as g++ or GCC. The system uses local text files for persistent storage and does not require an internet connection.

2.5 Constraints

The system is implemented as a console-based C++ application. Data is stored using local files rather than a centralized database. The system does not connect to real railway reservation servers, payment gateways, or real-time railway APIs. Train and fare information are simulated for academic purposes.

3. External interface requirements

3.1 User interfaces

Primary UI: A menu-driven console interface with clear options for searching trains, booking tickets, viewing bookings, and cancelling tickets.

Accessibility: Simple text-based prompts, clear messages, and easy-to-read menus for users with basic computer knowledge.

3.2 Hardware interfaces

The system requires a standard computer with keyboard, display, processor, and sufficient storage for application files.

3.3 Software interfaces

The system will use the C++ Standard Library and file input/output facilities for data processing and persistent storage.

3.4 Communications

No external network communication is required. All train, passenger, and booking information is stored and processed locally.

4. System features (detailed)

4.1 Train Search

Req ID	Requirement	Type	Priority	Stakeholder	Acceptance criteria	Test
RT-F001	The system shall display all available trains with their train number, name, source, destination, departure time, arrival time, fare, and seat availability.	Functional	High	Passenger	All stored train records are displayed with the complete set of listed fields.	TC-TRAIN-01
RT-F002	The system shall allow the user to search for trains using source and destination.	Functional	High	Passenger	Trains matching the given source and destination are displayed correctly.	TC-TRAIN-02
RT-F003	The system shall allow the user to search for a train using its train number.	Functional	High	Passenger	The system displays the correct train details when a valid train number is entered.	TC-TRAIN-03
RT-F004	The system shall display the number of available seats for a selected train.	Functional	High	Passenger	The system displays the correct number of available seats for the selected train.	TC-TRAIN-04

4.2 Ticket Booking

Passenger details shall include name, age, gender, and contact number.

Req ID	Requirement	Type	Priority	Stakeholder	Acceptance criteria	Test
RT-F005	The system shall allow the user to select a train and enter passenger details (name, age, gender, and contact number) for booking.	Functional	High	Passenger	User can select a valid train and enter the required passenger details.	TC-BOOK-01
RT-F006	The system shall validate the passenger details (name, age, gender, and contact number) before confirming the booking.	Functional	High	Passenger	Booking is accepted only when all required passenger details are valid.	TC-BOOK-02
RT-F007	The system shall check seat availability before confirming a booking.	Functional	High	Passenger	Booking is allowed only when at least one seat is available.	TC-BOOK-03
RT-F008	The system shall assign an available seat to the passenger after successful booking.	Functional	High	Passenger	A valid booking receives one available seat number.	TC-BOOK-04
RT-F009	The system shall generate a unique PNR for every confirmed booking.	Functional	High	Passenger	Each confirmed booking has a unique PNR.	TC-BOOK-05
RT-F010	The system shall calculate and display the ticket fare using the predefined fare associated with the selected train.	Functional	High	Passenger	The correct fare is displayed for the selected train and booking.	TC-BOOK-06
RT-F011	The system shall display booking confirmation after a successful booking.	Functional	High	Passenger	Confirmation displays PNR, passenger details, train, seat, date, and fare.	TC-BOOK-07

4.3 Ticket Cancellation

Req ID	Requirement	Type	Priority	Stakeholder	Acceptance criteria	Test
RT-F012	The system shall allow the user to search for a booking using the PNR.	Functional	High	Passenger	A valid PNR displays the corresponding booking.	TC-CAN-01
RT-F013	The system shall display booking details before cancellation.	Functional	Medium	Passenger	The correct booking details are displayed before cancellation.	TC-CAN-02
RT-F014	The system shall cancel a valid booking after user confirmation.	Functional	High	Passenger	A confirmed booking changes to cancelled after confirmation.	TC-CAN-03
RT-F015	The system shall make the cancelled seat available for future bookings.	Functional	High	Passenger	The cancelled seat is marked as available after cancellation.	TC-CAN-04
RT-F016	The system shall update the booking status to "CANCELLED".	Functional	High	Passenger	The booking status changes from confirmed to cancelled.	TC-CAN-05
RT-F017	The system shall calculate and display the refund amount after cancellation, based on the applicable cancellation policy.	Functional	Medium	Passenger	The displayed refund amount matches the defined cancellation policy.	TC-CAN-06

4.4 Booking Management

Req ID	Requirement	Type	Priority	Stakeholder	Acceptance Criteria	Test
RT-F018	The system shall allow users to view the details of an existing booking using the PNR.	Functional	High	Passenger	A valid PNR displays the correct booking details.	TC-MGT-01
RT-F019	The system shall prevent cancellation of an already cancelled booking.	Functional	High	Passenger	An already cancelled booking cannot be cancelled again.	TC-MGT-02
RT-F020	The system shall maintain booking records in persistent storage.	Functional	High	Passenger / Admin	Booking records remain available after the application is restarted.	TC-MGT-03

4.5 Administration & Data management

Req ID	Requirement	Type	Priority	Stakeholder	Acceptance Criteria	Test
RT-F021	The administrator shall be able to add new train information.	Functional	Medium	Administrator	A valid new train record is added and can be viewed in the system.	TC-ADMIN01
RT-F022	The administrator shall be able to modify existing train information.	Functional	Medium	Administrator	Updated train information is saved and displayed correctly.	TC-ADMIN02
RT-F023	The administrator shall be able to view all train and booking records.	Functional	Medium	Administrator	All stored train and booking records are displayed correctly.	TC-ADMIN03
RT-F024	The administrator shall authenticate (log in) before accessing any administrative function.	Functional	High	Administrator	Administrative menus are inaccessible until valid administrator credentials are entered.	TC-ADMIN04
RT-F025	The administrator shall be able to delete an existing train record.	Functional	Medium	Administrator	A deleted train record no longer appears in train search or listing.	TC-ADMIN05

5. Non-functional requirements (detailed)

The following non-functional requirements define measurable quality expectations for performance, reliability, usability, persistence, and maintainability of the Railway Ticket Reservation System.

Req ID	Requirement	Category	Priority	Stakeholder	Acceptance criteria / Test case
RT-NF001	The system shall display the result of a train search within 2 seconds for datasets containing up to 10,000 train records.	Performance	High	System User	Search results are displayed within 2 seconds in the defined test dataset. Test: TCNFR-01
RT-NF002	The system shall correctly update seat availability after every successful booking and cancellation.	Reliability	High	Passenger / Administrator	Seat count before and after booking/cancellation is consistent. Test: TC-NFR-02
RT-NF003	The system shall provide a simple menu-driven interface that can be used by a user with basic computer knowledge.	Usability	Medium	Passenger	A user can complete a basic booking flow without external instructions. Test: TC-NFR-03
RT-NF004	Confirmed and cancelled booking records shall remain available after the application is closed and restarted.	Data Persistence	High	Passenger / Administrator	Saved records are correctly loaded after restart. Test: TCNFR-04
RT-NF005	The system shall be divided into logically independent components for train, passenger, ticket, and reservation management.	Maintainability	Medium	Developer	Changes to one component can be made without changing unrelated components. Test: TC-NFR-05

5.1 Security

5.1.1 Security Objectives

- Security Objective 1: Prevent unauthorized modification of train and booking records.
- Security Objective 2: Protect passenger and booking information from unauthorized access within the application.

5.1.2 Security Requirements

Req ID	Requirement	Type	Priority	Acceptance criteria / Test case
PRJ-SR001	The system shall validate all user input before processing it.	Security	High	Invalid input is rejected without crashing the application. Test: TC-SEC-01
PRJ-SR002	The system shall reject invalid train numbers and PNR values.	Security	High	Invalid identifiers produce an error message and no record is modified. Test: TC-SEC-02
PRJ-SR003	The system shall prevent duplicate PNR numbers for different bookings.	Security	High	No two active bookings contain the same PNR. Test: TC-SEC-03
PRJ-SR004	The system shall prevent booking when no seats are available.	Security	High	Booking is rejected when the selected train has zero available seats. Test: TC-SEC-04
PRJ-SR005	Administrative functions shall require administrator authentication.	Security	High	Unauthorized users cannot access administrator functions. Test: TC-SEC-05

6. Quality attributes & Acceptance tests

The system will be accepted when the major functional requirements are implemented and verified, the defined non-functional requirements are met, and the planned test cases pass.

6.1 Quality Attributes

Quality Attribute	Expected Quality	Verification
Performance	Train searches complete within the specified response time.	TC-NFR-01
Reliability	Seat availability remains consistent after booking and cancellation.	TC-NFR-02
Usability	Users can navigate the menu and complete common operations with clear prompts.	TC-NFR-03
Data Persistence	Booking data remains available after application restart.	TC-NFR-04
Maintainability	System functions are organized into logical modules/classes.	TC-NFR-05
Security	Input validation, PNR uniqueness, seat checks, and administrator authentication are enforced.	TC-SEC-01 to TCSEC-05

6.2 Acceptance Test Suites

Test Suite	Purpose	Test Cases
Train Search	Verify display of trains, search by source/destination, search by train number, and seat availability.	TC-TRAIN-01 to TC-TRAIN-04
Ticket Booking	Verify passenger input, availability checks, seat assignment, PNR generation, fare calculation, and confirmation.	TC-BOOK-01 to TC-BOOK-07
Ticket Cancellation	Verify PNR lookup, details display, cancellation, seat release, status update, and refund display.	TC-CAN-01 to TC-CAN-06
Booking Management	Verify booking retrieval, prevention of duplicate cancellation, and persistent storage.	TC-MGT-01 to TC-MGT-03
Administration	Verify administrator login and train/booking management functions.	TC-ADMIN-01 to TC-ADMIN-05
Performance	Verify search response time on the defined dataset.	TC-NFR-01
Security	Verify input validation, invalid identifiers, PNR uniqueness, no-seat booking prevention, and admin authentication.	TC-SEC-01 to TC-SEC-05

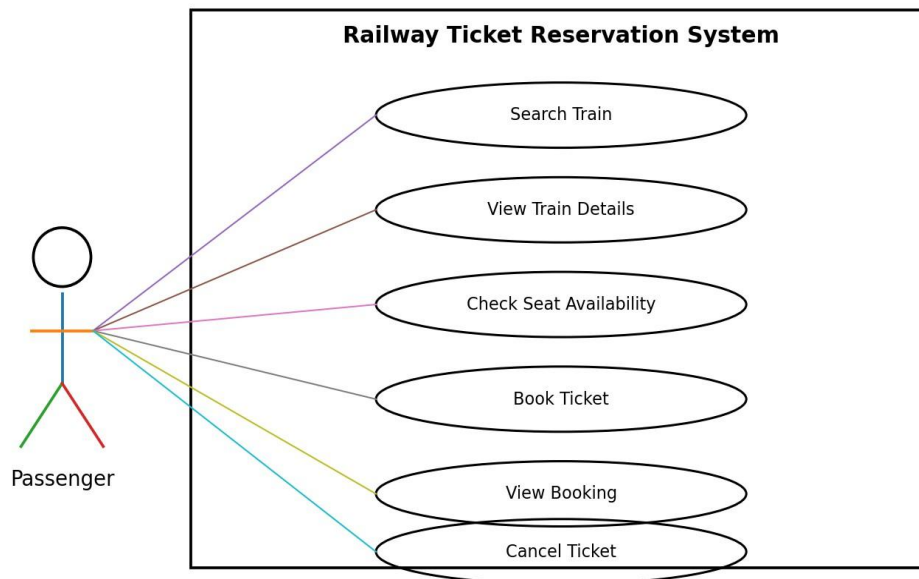
6.3 Exit Criteria

- All high-priority functional requirements are implemented and verified.
- No critical functional, security, or data-persistence failures remain.
- All planned acceptance test cases pass, or any failed test has a documented resolution/exception approved by the evaluator.
- The Requirements Traceability Matrix is completed and shows coverage of requirements by design modules and test cases.

7. System models and diagrams

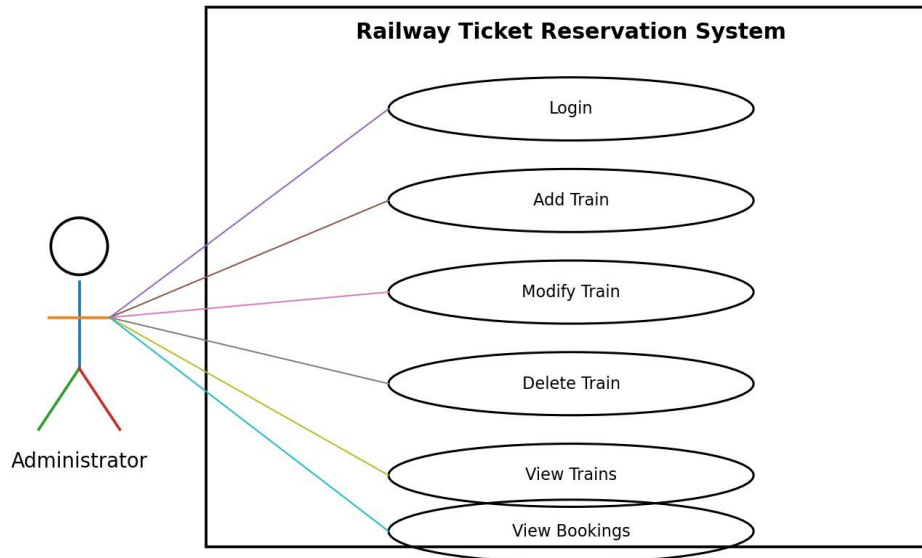
7.1 UML Use-Case Diagram 1 — Passenger

The Passenger interacts with the system to search trains, view train information, check seat availability, book tickets, view existing bookings, and cancel tickets.



7.2 UML Use-Case Diagram 2 — Administrator

The Administrator logs in and manages train and booking information through administrative operations.



7.3 Use-Case Summary

Actor	Use Case	Description
Passenger	Search Train	Find trains using source and destination or train number.
Passenger	View Train Details	View train number, name, route, timing, fare, and availability.
Passenger	Check Seat Availability	View currently available seats for a selected train.
Passenger	Book Ticket	Enter passenger details and reserve an available seat.
Passenger	View Booking	Retrieve booking details using the PNR.
Passenger	Cancel Ticket	Cancel a valid booking and release its seat.
Administrator	Login	Authenticate before accessing administrative operations.
Administrator	Add Train	Add new train information to the system.
Administrator	Modify Train	Update existing train information.
Administrator	Delete Train	Remove a train record when permitted.
Administrator	View Trains	View stored train records.
Administrator	View Bookings	View stored booking records.

8. Requirements Traceability Matrix (RTM)

The RTM maps every requirement to its SRS section, implementation/design module, and verification test case. Status values are N = Not implemented, P = Partially implemented, and A = Accepted.

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)
RT-F-001	Display all available trains	4.1 / DS-Train-01	TrainModule	TC-TRAIN-01	N
RT-F-002	Search by source and destination	4.1 / DS-Train-02	SearchModule	TC-TRAIN-02	N
RT-F-003	Search by train number	4.1 / DS-Train-03	SearchModule	TC-TRAIN-03	N
RT-F-004	Display available seats	4.1 / DS-Train-04	SeatModule	TC-TRAIN-04	N
RT-F-005	Select train and enter passenger details	4.2 / DS-Booking-01	BookingModule	TC-BOOK-01	N
RT-F-006	Validate passenger details	4.2 / DS-Booking-02	ValidationModule	TC-BOOK-02	N
RT-F-007	Check seat availability before booking	4.2 / DS-Booking-03	SeatModule	TC-BOOK-03	N
RT-F-008	Assign an available seat	4.2 / DS-Booking-04	SeatModule	TC-BOOK-04	N
RT-F-009	Generate unique PNR	4.2 / DS-Booking-05	TicketModule	TC-BOOK-05	N
RT-F-010	Calculate and display fare	4.2 / DS-Booking-06	FareModule	TC-BOOK-06	N
RT-F-011	Display booking confirmation	4.2 / DS-Booking-07	BookingModule	TC-BOOK-07	N
RT-F-012	Search booking using PNR	4.3 / DS-Cancel-01	BookingModule	TC-CAN-01	N
RT-F-013	Display booking details before cancellation	4.3 / DS-Cancel-02	BookingModule	TC-CAN-02	N
RT-F-014	Cancel valid booking after confirmation	4.3 / DS-Cancel-03	CancellationModule	TC-CAN-03	N
RT-F-015	Make cancelled seat available	4.3 / DS-Cancel-04	SeatModule	TC-CAN-04	N
RT-F-016	Update booking status to CANCELLED	4.3 / DS-Cancel-05	TicketModule	TC-CAN-05	N
RT-F-017	Display refund amount	4.3 / DS-Cancel-06	RefundModule	TC-CAN-06	N
RT-F-018	View booking details using PNR	4.4 / DS-BookingMgmt-01	BookingModule	TC-MGT-01	N

RT-F-019	Prevent duplicate cancellation	4.4 / DS-BookingMgmt-02	CancellationModule	TC-MGT-02	N
RT-F-020	Maintain persistent booking records	4.4 / DS-BookingMgmt-03	FileModule	TC-MGT-03	N
RT-F-021	Add new train information	4.5 / DS-Admin-01	AdminModule	TC-ADMIN-01	N
RT-F-022	Modify existing train information	4.5 / DS-Admin-02	AdminModule	TC-ADMIN-02	N
RT-F-023	View all train and booking records	4.5 / DS-Admin-03	AdminModule	TC-ADMIN-03	N
RT-F-024	Administrator authentication (login)	4.5 / DS-Admin-04	AdminModule	TC-ADMIN-04	N
RT-F-025	Delete a train record	4.5 / DS-Admin-05	AdminModule	TC-ADMIN-05	N
RT-NF-001	Train search response time	5 / DS-Perf-01	SearchModule	TC-NFR-01	N
RT-NF-002	Consistent seat updates	5 / DS-Reliability-01	SeatModule	TC-NFR-02	N
RT-NF-003	Simple menu-driven usability	5 / DS-UX-01	UserInterface	TC-NFR-03	N
RT-NF-004	Persistent records after restart	5 / DS-Storage-01	FileModule	TC-NFR-04	N
RT-NF-005	Logical component separation	5 / DS-Architecture-01	SystemArchitecture	TC-NFR-05	N
PRJ-SR001	Validate user input	5.1 / DS-Sec-01	ValidationModule	TC-SEC-01	N
PRJ-SR002	Reject invalid train numbers and PNRs	5.1 / DS-Sec-02	ValidationModule	TC-SEC-02	N
PRJ-SR003	Prevent duplicate PNRs	5.1 / DS-Sec-03	TicketModule	TC-SEC-03	N
PRJ-SR004	Prevent booking with no seats	5.1 / DS-Sec-04	SeatModule	TC-SEC-04	N
PRJ-SR005	Require administrator authentication	5.1 / DS-Sec-05	AdminModule	TC-SEC-05	N